

Efficient, Comfortable Homes Make Satisfied Buyers

Rising energy prices, coupled with increased public awareness, have fueled demand for more energy-efficient homes. Many builders have responded by qualifying their new homes for the Energy Star label. Others have gone beyond Energy Star standards and qualified their new homes through Guaranteed Performance (GP) programs such as Environments For Living (EFL) and Performance Built that allow builders to guarantee energy savings and comfort (see “Guaranteed Performance”). But have these programs worked? Has energy consumption actually been reduced? And have homeowners been satisfied with the performance of these program homes? My colleagues at Advanced Energy and I addressed these issues in parallel studies published in early 2006.

The Phoenix Home Energy Efficiency Study, sponsored by EPA, analyzed data from 7,141 homes built by six different production building companies in Phoenix between 1994 and 2004. There were 3,336 baseline homes, 2,979 Energy Star homes, and 826 GP homes. Details on the physical design and construction of these homes

(such as HVAC ratings, window type, floor area, and volume) were obtained from local builders, utilities, and testing companies. Energy use for the period 1998–2004 was obtained from Arizona Public Service and Southwest Gas Corporation. Energy consumption data were statistically adjusted to take into account such factors as cooling degree-days, floor area, whether the house had gas or electric heat, and the presence of a swimming pool. More-detailed results of that study were recently published in *Home Energy* (see “Market Changes,” *HE*, Sept/Oct ’06, p. 30).

The Phoenix Homeowner Satisfaction Survey, sponsored jointly by DOE and Arizona-based partners, was a direct-mail survey of homeowners in Phoenix who had bought a new home within the past five years. The survey went to the same homes that were studied in the Phoenix Home Energy Efficiency Study—approximately 7,000 surveys were sent out. We received responses from 205 baseline homeowners, 255 Energy Star homeowners, and 235 GP homeowners. The survey compared the attitudes of the three categories of hom-

owner regarding their satisfaction with the level of comfort in their homes.

An Energy Snapshot

Since detailed results for the Phoenix Home Energy Efficiency Study have already been published in *Home Energy*, here are some highlights:

Energy Star homes consume less energy. In Phoenix, Energy Star homes are up to 16% more efficient than comparable baseline, or code-built homes when comparing energy intensity kW/ft². GP homes do even better, consuming up to 33% less than code-built homes and up to 20% less than comparable Energy Star homes for cooling. These figures are for the summer cooling load and were adjusted for the presence of a swimming pool.

The energy study highlighted a trend that we believe is of concern to those who design, implement, or utilize energy efficiency programs. Although Energy Star homes and GP homes outperform regular baseline homes with regard to heating and cooling and water heating, actual total energy use is increasing. For more on this problem, see “Roadblocks

Guaranteed Performance

Guaranteed Performance (GP) is a general label for homes that have guaranteed energy bills and comfort. Environments For Living (EFL) is the largest such program. Arizona Public Service sponsors Performance Built, a smaller program. The homes in the Phoenix study that were labeled as GP include homes from Performance Built and two similar programs, Engineered For Life and Environments For Living. Engineered For Life was started at Louisiana Pacific many

years ago. The primary program developers at Louisiana Pacific decided to leave in 2001 and started a similar program at Masco Contractor Services called Environments For Living. Both programs have similar standards.

The Environments For Living program is based on specifications originally developed through a collaboration of Advanced Energy, Louisiana Pacific, and DOE’s Building America program. EFL qualification is based on strict standards that require a builder to commit to improving their homes thermal envelope (high per-

formance windows, properly installed insulation, advanced framing) and HVAC equipment (tight ducts and high efficiency equipment). The homes’ energy bills can be predicted and occupant comfort controlled so that a builder can offer a guarantee on the homes’ performance. There are over 100,000 EFL-labeled homes in the United States with tens of thousands added every year.

For more on the Environments For Living program, go to www.eflhome.com.

to Zero-Energy Homes,” *HE*, Jan/Feb '07, p. 24.

Customer Satisfaction

The Phoenix Homeowner Satisfaction Survey found that performance isn't just about energy efficiency. Homeowners say it's also about comfort, reliability, and healthiness. Some compared it to the fuel economy of a car. If they're not going to be comfortable driving the car, why should they worry about fuel economy?

Guaranteed Performance homeowners are more satisfied on almost all the survey measures of satisfaction relating to comfort, healthiness, energy efficiency, and performance. GP homeowners are more satisfied with every aspect of their homes' HVAC performance compared to the other homeowners; 49% of GP homeowners said they were completely satisfied with their homes' "ability to keep them comfortable year-round" compared to 35% of Energy Star homeowners and only 27% of baseline homeowners.

When asked "how satisfied have you been with the freshness of the air inside your home during those times when you keep the doors and windows shut," 40% of GP homeowners were completely satisfied, compared to 24% of Energy Star homeowners and 20% of baseline homeowners. Other questions asked about the evenness of temperatures from room to room, reliability, and cooling cost.

The Satisfaction Survey found that buyers of new homes are generally unaware of energy programs. Marketers still have a way to go to build awareness of Energy Star Homes and GP programs. The survey found that 56% of homeowners were aware of the Energy Star program, but only 10% were aware of any guaranteed performance program, such as EFL. Surprisingly, 37% were not aware of any special energy program. These statistics all come from a market that was an early adopter of both the Energy Star Homes and GP programs.

Making the Business Case for Energy Efficiency

These studies have important implications for the new-home market in North America. Specifically, they begin to lay the basis for a business case for energy efficiency, where energy-efficient homes compete in a market-based environment. Instead of asking people to do something they prefer not to do, such as turn down their thermostats, these studies demonstrate that it's possible to build a home that homeowners prefer because it performs better, not just because it uses less energy.

These studies have implications for builders, utilities, and equipment manufacturers. For builders, the implication is that buyers will notice the difference in a program home. Energy efficiency can become a way for builders to differentiate themselves. For utilities, the implication is the possibility that their load profile can be improved without the need for continuing subsidies. And

for equipment manufacturers, the payoff is homeowners who are more satisfied with both the reliability and the cost of operation of their homes' heating and cooling systems.



—Colby Swanson

Colby Swanson is director of building consulting with Advanced Energy in Raleigh, North Carolina. The group works with production builders nationwide to reduce liabilities, callbacks, and claims, and to improve profits and customer satisfaction.

For more information:

The full reports on the Phoenix energy efficiency and homeowner satisfaction surveys are available on the Advanced Energy Web site, www.advancedenergy.org/buildings/services/research.html.